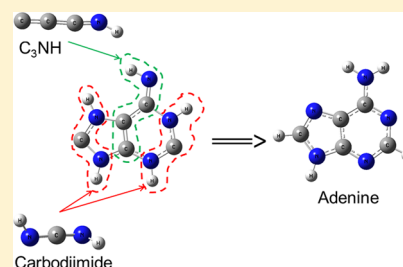


Adenine Formation without HCN

Kenneth M. Merz, Jr.,^{†,‡} Eduardo C. Aguiar,[§] and Joao Bosco P. da Silva^{*,§}[†]Quantum Theory Project, University of Florida, 2234 New Physics Building, Gainesville, 32611 Florida, United States[‡]Institute for Cyber Enabled Research, Department of Chemistry, and Department of Biochemistry and Molecular Biology, Michigan State University, 578 South Shaw Lane, East Lansing, Michigan 48824, United States[§]Departamento de Química Fundamental, Universidade Federal de Pernambuco, Recife, 54740-540 Pernambuco, Brazil

S Supporting Information

ABSTRACT: From a historic point of view adenine was always presumed to be the product of HCN pentamerization. In this work a new mechanism for adenine synthesis in the gas phase without HCN is proposed. The concept of retrosynthetic analysis was employed to create a tautomer of adenine, which can be reached from previously observed interstellar molecules C_3NH and $HNCNH$ and its isomer H_2NCN . MP2/6-311++G(2d,2p) calculations were performed to calculate the Gibbs free energy of the minimum and the transition state (TS) structures involved in the six step mechanism. This new mechanism requires a smaller number of steps, the reaction energy is twice as exergonic, and the rate determining TS is lower in energy than the corresponding ones proposed elsewhere in the literature.



1. INTRODUCTION

It is well-known that the DNA's nitrogen bases adenine, together with guanine, cytosine, and thymine are key molecules for living systems as we know them on Earth. Understanding how these molecules were naturally formed is a fundamental problem whose answer will link so disparate sciences as biochemistry and astronomy. Herein we analyze the formation of adenine from interstellar molecules with the aim of providing insight into how this important molecule is formed in the interstellar medium (ISM).

The pioneering experiments of Miller and Urey¹ examined the generation of organic molecules under experimental conditions aimed at mimicking the atmosphere of the primitive earth. Subsequent to this experimental effort others have investigated the synthesis of adenine in either aqueous^{2–5} media involving the presence of hydrogen cyanic acid (HCN) or ammonium cyanide (NH_4CN). The focus on HCN for the synthesis of adenine is easy to understand since its molecular formula is $H_5C_3N_5$ and HCN, as well its isomer CNH, are well documented species in many interstellar molecular clouds.^{6,7} A number of years ago, Matthews and Minard have stressed that HCN, in the presence of bases like ammonia or amines, may lead to the generation of two HCN polymers, heteropolyamides, which under hydrolysis conditions can be transformed into heteropolypeptides and polyimines, which by the addition of HCN and subsequent hydrolysis reactions may lead to polynucleobases and nucleobases, respectively.⁸

However, the presence of atmosphere favorable toward the synthesis of adenine is not exclusively a peculiarity of the Earth. In 1944, Kuiper proposed the presence of an atmosphere on Titan, the largest Saturn satellite, by the measurement of near-infrared absorption bands of methane.⁹ In 1980, the Titan flyby of Voyager 1 revealed the presence of other alkanes like ethane

and propane¹⁰ as well as HCN in its atmosphere.¹¹ More recently, in 2004, the Cassini–Huygens mission confirmed Titan's rich chemical repertoire by recording the infrared spectrum of many hydrocarbons containing nitrogen atoms and/or π -bond, e.g., H_2CCH_2 , $HCCH$, CH_3CCH , C_6H_6 , HCN, HC_3N , and C_2N_2 .^{12,13} As a consequence of this chemical inventory and light exposure (even being further from the Sun than the Earth is), the Titan haze or aerosol (observed by the Huygens probe when it landed on Titan's surface in 2005¹⁴) can be explained in terms of polymeric products also obtained in the laboratory when a mixture of N_2 , CH_4 , H_2O , and CO_2 is exposed to radiation sources at low temperatures. Thus, still within the spirit of the Miller–Urey experiments, Pilling and co-workers have reported the detection of adenine using GC/MS and 1H NMR techniques when Titan aerosol analogues were irradiated with soft X-ray photons.¹⁵ In another paper, these authors have also reported an interesting aspect about adenine survival in interstellar conditions, its higher stability to ionizing radiation when compared to other nitrogen bases.¹⁶ Finally, it is interesting to note that adenine has been reported in two different analyses of cometary materials, namely, from samples from the Murchison carbonaceous chondrite¹⁷ and during the Giotto mission for Halley's comet.¹⁸

Overall, the evidence of adenine formation on condensed medium that includes planetary/satellite biotas or cometary material is fairly clear, but to date adenine has not been observed in gas-phase molecular clouds even though evidence shows that adenine is stable under these conditions. For instance, Brown and co-authors have concluded from the

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comparison between calculated and experimental rotational constants that the N(9)H form of adenine is the most stable tautomer in the gas phase.¹⁹ Second, Michaelian and co-authors have found many times the “natural” configuration of adenine in global searches for stable adenine isomers using a genetic algorithm coupled with a force field.²⁰ Moreover, since the 1960s it has been known that adenine has the largest stabilizing resonance energy of all purine and pyrimidine nitrogen bases.²¹

Greenberg has discussed in detail the importance of comets as a delivery vehicle of organic molecules in biotas like the Earth.²² Since copper among²³ other transition metals²⁴ is found in carbonaceous chondrites, the observation of Furukawa, Tanaka, and Kawai²⁵ of the formation of low-dimensional superstructures (or super hydrogen-bond complexes) of adenine on Cu(111) surfaces supports the notion of comets providing an appropriate microenvironment to spread adenine in the universe.

Indeed, the amount of available adenine to contaminate bodies like planets, comets, and meteors has been estimated by Chakrabarti and Chakrabarti using kinetic models for the pentamerization (i.e., the fifth successive addition) of HCN in the gas phase.²⁶ However, Smith, Talbi, and Herbst using quantum chemistry calculations and kinetic arguments have concluded that the formation of adenine by HCN pentamerization is precluded by a 71 kcal mol⁻¹ energy barrier associated with the first chemical step, the formation of the dimer (H₂C₂N₂) formed from two HCN molecules.²⁷ Recently, the height of this activation energy was confirmed by Choe and Yim.²⁸

Nevertheless, in 2007 two independent works employing electronic structure calculation have investigated the mechanism of HCN pentamerization, focusing on the final step, i.e., the pyrimidine-ring formation by addition of the last HCN molecule to the amino-*H*-imidazole-carbonitriles (AICN) tetramer.^{29,30} While Glaser and co-authors,²⁹ using MP2-(full)/6-31G** calculations, have considered the compound formed by the addition of HCN to the tautomer 5-amino-1*H*-imidazole-4-carbonitrile molecule, Schleyer and co-authors,³⁰ using B3LYP/6-311+G** calculations, have examined the compound formed by the addition of HCN to the other tautomer (which they calculated as the most stable) 4-amino-1*H*-imidazole-5-carbonitrile as the starting point for the last step of the proposed mechanism of pentamerization of HCN to adenine. Regardless, of the differences in both the level of theory and starting point used, both groups have concluded that the neutral uncatalyzed reactions are kinetically prohibitive. To overcome this, Glaser and co-author have proposed photoactivation, whereas Schleyer and co-authors have proposed water or ammonia catalyzed processes.

Recently, Gupta and co-workers³¹ have used B3LYP/6-31G** calculation to investigate the formation of adenine by starting from species observed in the ISM, namely, HCN and HCCN considering radical–radical and radical–molecule interactions with other species like CN and H. The results of these authors point to the possibility of obtaining adenine following a mechanism where HCN reacts with several species via barrierless and exothermic processes.

Although there are many reasons to focus on the synthesis of adenine using HCN molecules, a natural and not yet properly explored question is it possible for adenine to be created in the interstellar environment though some neutral–neutral reactions without the involvement of HCN? If possible, from an energetic point of view, how does it compare with previously

calculated mechanisms? In order to answer these questions we present a new mechanism for adenine formation using a set of molecules previously observed in the ISM. The mechanism was inspired using a retrosynthetic analysis.³²

2. METHODOLOGY AND CALCULATIONS

2.1. Electronic Structure Calculation. Electronic structure calculations were carried out at the second-order Møller–Plesset Perturbation (MP2) level of theory³³ using the 6-311++G(2d,2p) basis set.³⁴ This level of theory was used throughout to determine the optimized geometries of all ground and transition state species involved in this study. The MP2 level of theory was selected because electron correlation has been demonstrated to be important for obtaining good geometries for nucleic acid bases.³⁵ The 6-311++G(2d,2p) basis set was chosen since diffuse functions have been shown to be important when considering intermolecular interactions³⁶ while also minimizing the basis set superposition error (BSSE) compared to smaller basis sets.³⁷ Vibrational harmonic frequency calculations were carried out in order to characterize stationary points on the potential energy surface and ground or transition states. Thermochemistry calculations were carried out using standard methods described elsewhere.³⁸ In this work, the computation of the Gibbs free energy was preferred since Kwiatkowski and Leszczynski have shown that enthalpy and entropy contributions may be important for the free tautomerization energies of DNA bases.³⁹ All calculations were performed using the Gaussian 09 program.⁴⁰ The frequency calculations were done at a temperature of 298.15 K and a pressure of 1 bar. The QST2 and QST3 methods⁴¹ were used to search for saddle points. The transition state was properly characterized (one and only one imaginary frequency), and intrinsic reaction coordinate (IRC) calculations⁴² were performed in order to check that the TS structures indeed connected the expected reactants and products. The hydrogen bond (H-bond) energy was calculated as the difference between the Gibbs free energy of the hydrogen bond complex (H-complex) minus the Gibbs free energy of the monomers.

2.2. Two Level Factorial Design. We employed a two level factorial design (TLFD) to create the tautomers of adenine for subsequent computation. Since in the nucleotide adenine is bound to ribose through the nitrogen atom 9 (N9), it is typical to have a hydrogen atom attached at that position (N9H) when one considers the isolated nucleobase.⁴³ As a consequence, only three domains on the adenine backbone remain for chemical modification. They are depicted in Figure 1.

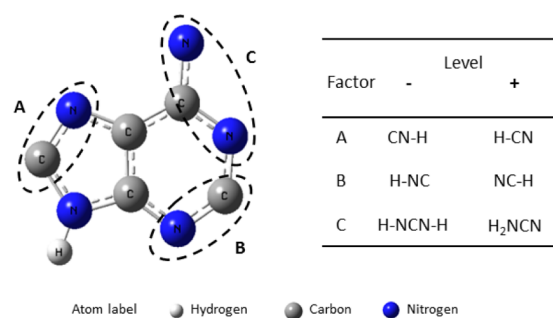


Figure 1. Factors that can influence the relative Gibbs free energy when changing the hydrogen position on the adenine N9H backbone structure considered at two levels (minus and plus).

These structural modifications shown in Figure 1 were not randomly selected but indeed they represent a full set of 2^3 theoretical experiments following a TLFD.⁴⁴ Using this chemometric multivariate exploratory technique it is possible to quantify the principal and interaction effects related to each factor that can influence the response function (in this case the relative Gibbs free energy). In Table 1 the codified design matrix corresponding to the structures of Figure 1 is presented.

Table 1. Codified Design Matrix for the 2^3 TLFD Analysis

structure	effect						
	principal			interaction			
	A	B	C	AB	AC	BC	ABC
1	–	–	–	+	+	+	–
2	+	–	–	–	–	+	+
3	–	+	–	–	+	–	+
4	+	+	–	+	–	–	–
5	–	–	+	+	–	–	+
6	+	–	+	–	+	–	–
7	–	+	+	–	–	+	–
8	+	+	+	+	+	+	+

In a quick inspection of Table 1 one observes that structure 8 represents adenine. Therefore, it is possible to calculate the relative standard Gibbs free energy (to adenine, i.e., $\Delta G_i^0 = G^0(i) - G^0(\text{adenine})$, $i = 1-8$). To evaluate the principal and interaction effects, the calculated ΔG^0 values are applied to the signs of the corresponding effect column in Table 1 and then the average value of the high level (+) minus the average value of the low level (–) is evaluated. For example, the principal effect C is given by the difference between the average value of the four ΔG^0 s calculated with the amine structures and the average value of the four ΔG^0 s calculated with the imine structures. Note that the sign for the interaction effects are obtained by the product of the signs of the principal effects (see Table 1).

3. RESULTS

3.1. Adenine Tautomers. Figure 2 shows the MP2/6-311++G(2d,2p) results for the structure, the lowest harmonic vibrational mode (ν_1), and the relative standard Gibbs free energy (ΔG^0) of the eight tautomers of adenine 1–8 (the 2^3 TLFD experiments) generated by changing the hydrogen position on the adenine N9H backbone structure (Figure 1).

From Figure 2 we observe that all tautomers of adenine represent energy minima. The principal and interaction effects on the ΔG^0 calculated for the tautomers of adenine are shown in Figure 3. In Figure 3a one can observe that the two principal effects A and B as well as the interaction effect BC are the most important effects on changing the hydrogen position on the adenine N9H backbone structure.

The interpretation of the principal effects is straightforward. The change of the hydrogen attachment position from the nitrogen to the carbon atoms in the five (A)- and the six-membered (B) rings decreases the ΔG^0 value, on average, by -32 and -46 kcal mol⁻¹, respectively. Otherwise, the effect C is the lowest principal effect. This means that the tautomerization from imine to amine in the C moiety decreases the Gibbs free energy less than that of the corresponding changes in A and B. The interpretation of the largest interaction effect BC comes from Figure 3b. It means that the change in C from the minus

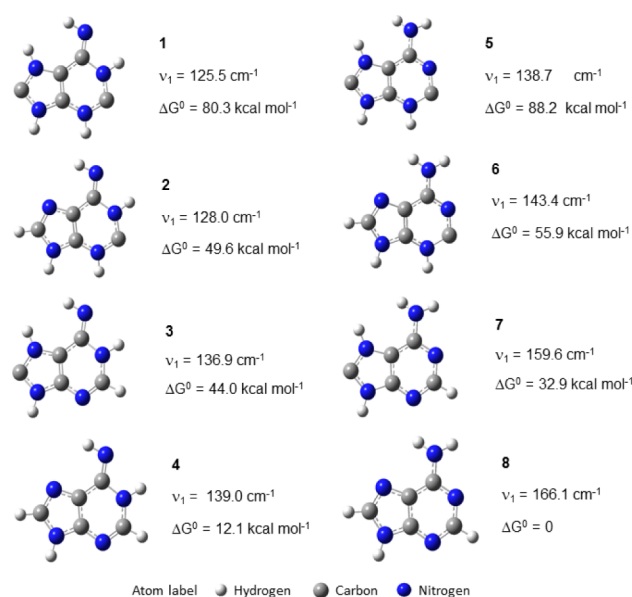


Figure 2. MP2/6-311++G(2d,2p) relative standard Gibbs free energy and the lowest harmonic vibrational mode of the imine (1–4) and amine (5–8) tautomers of adenine. $G^0(8) = -466.194652$ e_H.

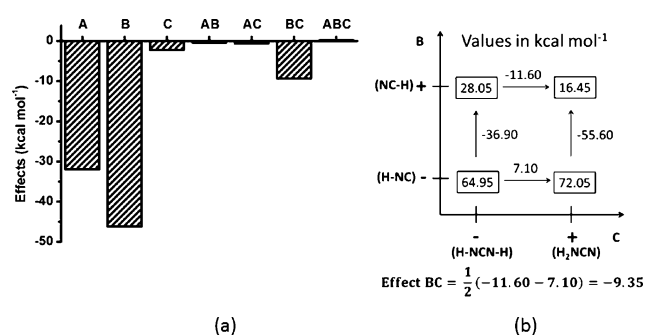


Figure 3. (a) Principal and interaction effects on the structural modification of the tautomers of adenine. (b) Interaction effect BC.

(H–NCN–H) to the plus (H₂NCN) level at the high level of B is exergonic (-11.60 kcal mol⁻¹), whereas the same change at the low level of B is a nonspontaneously endergonic (7.10 kcal mol⁻¹) process.

3.2. Retrosynthetic Analysis. The concept of retrosynthetic analysis was employed to breakdown tautomer 8 (Figure 2) because it is possible to directly relate 1 to previously observed molecules in the ISM C₃NH and HNCNH (see Figure 4).

Because of the low concentration conditions found in the ISM (even inside of the so-called dense molecular clouds),⁴⁵ it is statistically appropriate to neglect a three-body collision reaction between these species. Therefore, we propose only bimolecular reactions toward 1 involving first the formation of the six-membered ring followed by the fusion of the five-membered ring. From 1 to 8 there are many possible tautomerization sequences. The TLFD's effect values (Figure 3) guided us through an exergonic pathway. Because the interaction effect AB is small, the first tautomerization may occur at moiety B (the largest principal effect) even if the five-membered ring has not been created yet. The second principal effect A indicates that the next tautomerization process will occur at the moiety A. Finally, because the interaction effect BC is not negligible, the tautomerization from the imine to the

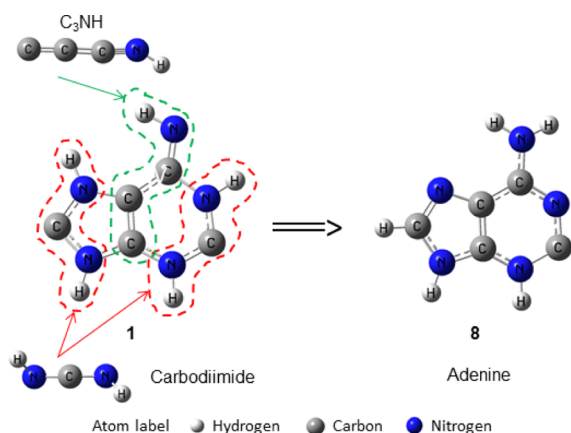


Figure 4. Retrosynthetic analysis for the formation of imine form (1) from the observed interstellar molecules HNCNH and C₃NH.

amine will occur only when the tautomerization (H–NC → NC–H) in the moiety B has occurred.

3.3. Formation of the Six-Membered Ring. The minima and the corresponding transition states for the formation of the six-membered ring 12 are shown in Figure 5. The related energetic, vibrational, and polarity parameters for these structures and transition states are shown in Table 2.

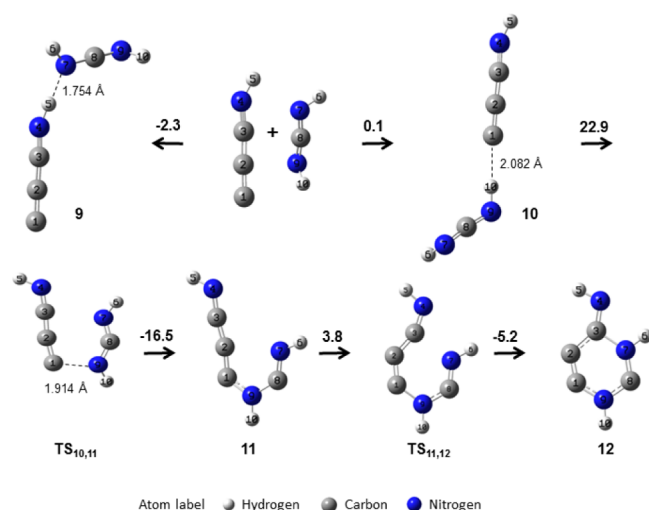


Figure 5. MP2/6-311++G(2d,2p) structures and standard Gibbs free energy, in kcal mol^{−1}, involved in the formation of the six-membered ring 12 involving C₃NH and HNCNH.

Table 2. Standard Gibbs Free Energy (G^0), H-Bond Energy ($\Delta_{\text{H-bond}}G^0$), Lowest Vibrational Frequency (ν_1), and Dipole Moment (μ) for the Minima and Transition State for the Species in Figure 5

structure	G^0 (e _H)	$\Delta_{\text{H-bond}}G^0$ (kcal mol ^{−1})	ν_1 (cm ^{−1})	μ (D)
HNCNH	−148.447 783		535.3	2.03
C ₃ NH	−169.093 609		157.8	6.38
9	−317.545 110	−2.33	11.8	10.43
10	−317.541 210	0.11	22.9	9.57
TS _{10,11}	−317.504 695		−448.5	3.03
11	−317.531 008		77.7	0.42
TS _{11,12}	−317.524 906		−331.0	1.43
12	−317.533 217		164.7	1.84

From Table 2 we find that each structure (9–12) represents a minimum on the potential energy surface. The potential of C₃NH for making H-bond complexes noted elsewhere^{46,47} is observed again when it interacts with HNCNH (H-complexes 9 and 10 in Figure 5). The large electric dipole moment for 9 (10.43 D) and 10 (9.57 D) makes them candidates to observation in the rotational domain. The quality of our prediction for the electric dipole moment may be appreciated by the comparison between the calculated and experimental values for HNCNH, 2.03 D versus 1.90(5) D, respectively.⁴⁸ Except the TS_{9,10}, all the other TSs were confirmed by the IRC calculations. Because of that, the H-complex 10 was considered the first organized structure toward the formation of the six-membered ring 12. The formation of a covalent bond, which forms 11, represents the rate-determining step (barrier of 23 kcal mol^{−1}) for the formation of the six-membered ring of adenine. Wang and Bowie have predicted through B3LYP/6-311++G(d,p) calculations that the reaction between C₃NH and NH₂ of urea has a barrier of 32 kcal mol^{−1} leading to an intermediate (closely related structure to 12) in a proposed mechanism for the formation of cytosine.⁴⁹ Following, the six-membered ring 12 is easily obtained in the *E* form after a minor C1C2C3 fragment reorientation associated with an energy barrier of only 3.8 kcal mol^{−1}. Our calculations predict the (*E*)-12 isomer to be more stable than (*Z*)-12 by 3.7 kcal mol^{−1}. The interconversion of (*E*)-12 to (*Z*)-12 is predicted to have a barrier height of 18.3 kcal mol^{−1}. These results are in good agreement with those predicted at the MP2(full)/6-31G(d) level of theory by Glaser and co-workers for the closely related system pyrimidin-4(3*H*)-imine (2.8 and 25.6 kcal mol^{−1}, respectively).⁵⁰ The stabilization due to the formation of the six-membered ring plays a dual role. First, it increases the terminal carbon atoms reactivity on the fragment C1C2 with important implications for the formation of the fused five-membered ring as will be discussed on the next section. Second, it increases the acidity of the N–H bonds (as can be observed by the N–H stretching frequency red shifts of −35, −24, and −11 cm^{−1}) making them more available to make intermolecular H-bonds. Although the formation of the covalently bonded structures 11 and 12 changes the infrared profile of the N–H oscillators, it leads to low electric dipole moment values of 0.42 D (11) and 1.84 D (12), which makes the rotational detection of these species a more challenging task.

3.4. Formation of the Double Fused Rings 6. From Figure 2 we find that for the imine structures the change on going from 2 to 4 is −37.5 kcal mol^{−1}, while from 1 to 3 it is −36.3 kcal mol^{−1}. Therefore, after creation of 12 the transformation from 13 to 14 represents the next natural step toward the synthesis of adenine (Figure 6). The energetic, vibrational, and electric dipole parameter values for structures 13–15 and 3 and the related transition states are shown in Table 3.

In this table structures 13–15 and 3 represent minima on the potential energy surface. Although proton tunneling processes have been evoked to explain the overcoming of high energy barriers for chemical reactions⁵¹ and physical processes⁵² of interest in the ISM, the transformation from 13 to 14 assisted by a carbodiimide (HNCNH) molecule (see Figure 6) represents a more efficient process because it decreases the energy barrier from 43.2 to 7.4 kcal mol^{−1} and increases the product (15) stability from −25.1 kcal mol^{−1} to −30.8 kcal mol^{−1} (see Table 2). Besides, the transformation from 13 to 14 should lead to the isomerization of HNCNH to H₂NCN, which

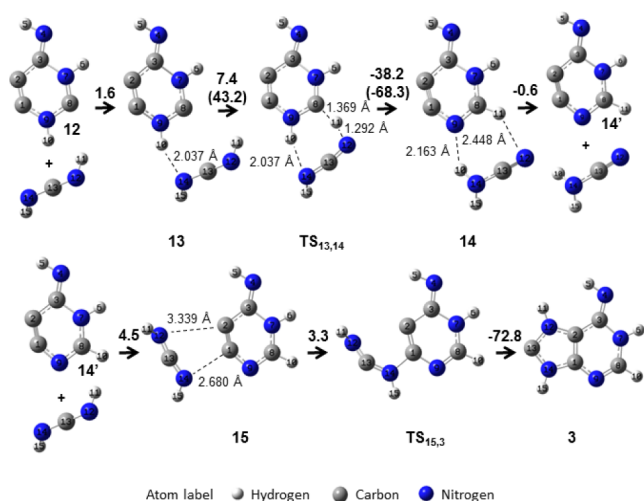


Figure 6. Formation of fused five-membered ring 3 from 12 and carbodiimide (HNCNH). Standard Gibbs free energies in kcal mol⁻¹. Values in parentheses are referenced to direct hydrogen transfer.

Table 3. Standard Gibbs Free Energy (G^0), H-Bond Energy ($\Delta_{\text{H-bond}}G^0$), Lowest Vibrational Frequency (ν_1), and Dipole Moment (μ) for the Minima and Transition State for the Species in Figure 6

structure	G^0 (ϵ_{H})	$\Delta_{\text{H-bond}}G^0$ (kcal mol ⁻¹)	ν_1 (cm ⁻¹)	μ (D)
H ₂ NCN	-148.455 409		397.8	4.41
13	-465.978 514	1.56	36.7	2.03
TS _{13,14}	-465.966 776		-1201.8	4.73
TS _{12,14'}	-317.464 366		-1818.9	1.14
14	-466.027 676	0.59	32.0	2.37
14'	-317.573 212		162.5	2.14
15	-466.013 787		17.2	1.75
TS _{15,3}	-466.008 521		-283.9	5.88
3	-466.124 461		136.9	2.63

is another well-established interstellar species.⁵³ Indeed, Borguet and co-workers have highlighted the role of H-bonds on the reverse isomerization process, i.e., the conversion from H₂NCN to HNCNH on ice surface.⁵⁴ In fact, the ability of H₂NCN to form H-bond interactions was first shown by King and Strope through infrared studies.⁵⁵

Since enyne-carbodiimide environments have been reported to form five-membered ring structures like imidazole through a Diels-Alder reaction,⁵⁶ we believe that the corresponding enyne form 14' can directly form 3. This step is predicted to have a low energy barrier of 4.5 + 3.3 kcal mol⁻¹ to reach 3, in a highly exergonic process (-65.0 kcal mol⁻¹). Fabian using B3LY/6-311+G(d,p) calculations predicted the barrier and the reaction Gibbs free energies for the addition of HNCNH to acetylene in 36.3 and -32.1 kcal mol⁻¹, respectively.⁵⁷ We have repeated the addition of HNCNH to acetylene, but using MP2/6-311+G(d,p) calculations, and the corresponding ΔE values were 29.1 and -36.5 kcal mol⁻¹, whereas the ΔG energetic parameters obtained were 35.9 and -19.1 kcal mol⁻¹, respectively. Therefore, the smaller barrier and larger stability on the transformation from 15 to 3 (relative to the addition of HNCNH to acetylene) is not due to the differences between the MP2 and B3LYP methods, but due to the peculiar CC moiety in 14'. While our MP2/6-311++G(2d,2p) calculations predict the CC bond distance for acetylene and ethylene to be 1.211 and 1.334 Å, respectively, we predict an intermediate

value for 14' of 1.245 Å. The triple bond in the cyclic species 4,5-pyrimidine (a closely related structure to 14') was predicted by the MNDO method to be 1.27 Å.⁵⁸ We find that the Diels-Alder reaction takes place through a non-synchronous low energy TS_{15,3} toward a stable and delocalized double ring system 3. This observation is not surprising given the intermediate nature of the CC bond (double/triple bond), which then reacts with an unsymmetrical ring moiety 14' to form 15 and ultimately 3. Another example of a Diels-Alder reaction involving a cyclic triple bond species like 14' was described by Promel and co-authors who studied the trapping 2-*t*-butyl-4,5-pyrimidine by furan.⁵⁹

3.5. Tautomerization toward Adenine Formation. The minima and the corresponding transition states for the formation of adenine are shown in Figure 7. The related energetic, vibrational, and electric dipole parameters for these structures and transition states are shown in Table 4.

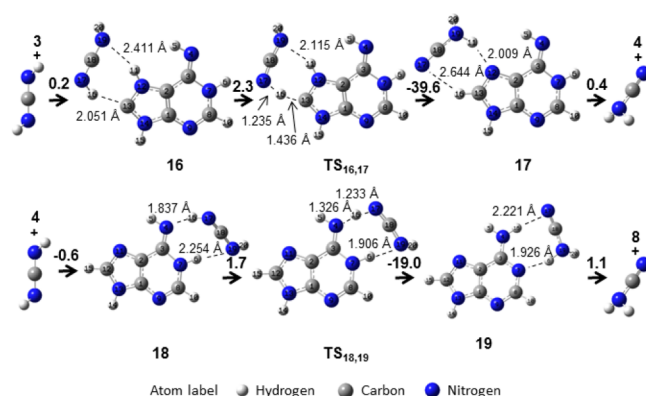


Figure 7. Tautomerization process assisted by HNCNH or NCNH₂ hydrogen bonded complexes toward adenine formation. Standard Gibbs free energies in kcal mol⁻¹.

Table 4. Standard Gibbs Free Energy (G^0), H-Bond Energy ($\Delta_{\text{H-bond}}G^0$), Lowest Vibrational Frequency (ν_1), and Dipole Moment (μ) for the Minima and Transition State for the Species in Figure 7

structure	G^0 (ϵ_{H})	$\Delta_{\text{H-bond}}G^0$ (kcal mol ⁻¹)	ν_1 (cm ⁻¹)	μ (D)
3 + HNCNH	-614.572 244			
16	-614.571 944	0.19	40.2	4.33
TS _{16,17}	-614.568 349		-1066.4	6.89
17	-614.631 417	-0.40	22.9	4.40
4 + HNCNH	-614.623 154			
18	-614.624 145	-0.60	33.4	4.89
TS _{18,19}	-614.621 430		-974.0	7.16
19	-614.651 754	-1.06	35.8	5.10
8 + H ₂ NCN	-614.650 061			

Again in this table all structures are minima on the potential energy surface. The energetic improvement on going from 16 to 17 (i.e., a lower expenditure of energy to overcome the transition state, 2.3 kcal mol⁻¹, and higher exergonic character of the reaction, -37.3 kcal mol⁻¹) compared to the similar process from 13 to 14 may be credited to the larger electron delocalization capabilities of 3 relative to 12. Finally, adenine can be obtained from the H-bonded complex 19 via 18 after overcoming a low energy barrier of 1.7 kcal mol⁻¹ and the release of -17.3 kcal mol⁻¹ of energy. Observe that a lower

reaction energy without the assistance of HNCNH/H₂NCN is predicted by our MP2/6-311++G(2d,2p) results $\Delta G = -12.1$ kcal mol⁻¹ (see 4 → 8 in Figure 1), $\Delta G = -12.4$ kcal mol⁻¹ at MP2(full)/6-31G(d,p) level of theory,²⁹ and $\Delta E = -12.3$ kcal mol⁻¹ at MP2/6-31G(d,p) level of theory.⁶⁰ However, the height of the energy barrier for the last transformation (4 → 8) when assisted by two water or two ammonia molecules are predicted, at B3LY/6-311+G(d,p) level of theory, to be 7.5 and 10.9 kcal mol⁻¹, respectively.³⁰

4. CONCLUDING REMARKS

On the basis of the chemical concept of retrosynthetic analysis, a new mechanism for adenine formation in the gas-phase has been proposed based on three known interstellar molecules: C₃NH and the isomers HNCNH and H₂NCN. This new mechanism for adenine formation in the gas-phase is very distinct to previously described HCN pentamerization mechanisms.^{29,30} First of all, it does not involve HCN at all, and only bimolecular steps are involved without the need for water or ammonia. Second, the entire proposed mechanism involves just 6 steps, while the four stages of HCN pentamerization assisted with one water molecule involves at least 14 steps, which can be decreased to 7 steps when assisted by two water molecules and excluding configurational and conformational changes.³⁰ Third, our mechanism proposes first the formation of the six-membered heterocyclic ring **12** and then the formation of the five-membered ring in **3**. Fourth, the NH/CH tautomerization assisted by one HNCNH molecule is a more facile process compared to the one assisted by two NH₃ or two H₂O molecules³⁰ from an energetic point of view (it leads to the formation of the most stable isomer, H₂NCN).

Importantly, the energetic parameters of our mechanism are more favorable than those obtained for alternative pathways. Our highest energy barrier, 22.9 kcal mol⁻¹ (TS_{10,11}), is much lower than the 71 kcal mol⁻¹ associated with the barrier for HCN dimer formation (H₂C₂N₂) or even the range of barriers of 27.6 kcal mol⁻¹ up to 37.6 kcal mol⁻¹ (using different models and levels of theory) for the addition of HCN to the HCN tetramer (AICN).³⁰ The subsequent ones involve, in general, barriers of decreasing heights: 3.8 kcal mol⁻¹ (TS_{11,12}), 7.4 kcal mol⁻¹ (TS_{13,14}), 3.3 kcal mol⁻¹ (TS_{15,3}), 2.3 kcal mol⁻¹ (TS_{16,17}), and 1.7 kcal mol⁻¹ (TS_{18,19}). Furthermore, as the mechanism moves toward adenine formation, the NH/CH tautomerization process over the aromatic heterocyclic ring(s) assisted by hydrogen bonded complexes involving HNCNH or H₂NCN becomes an increasingly spontaneous process because the $\Delta_{\text{H-bond}}G$ decreases from 1.56 kcal mol⁻¹ (**13**) to 0.59 kcal mol⁻¹ (**14**), 0.19 kcal mol⁻¹ (**16**), -0.40 kcal mol⁻¹ (**17**), -0.60 kcal mol⁻¹ (**18**), and -1.06 kcal mol⁻¹ (**19**). Thus, the overall Gibbs free energy of reaction is -142.2 kcal mol⁻¹ at 298.15 K and 1 bar, whereas it was predicted to be -53.7 kcal mol⁻¹ for the HCN pentamerization reaction.³⁰

Finally, there are known processes where the reaction products, having deep potential energy minimum, can lack an efficient channel to release this energy or to redistribute the vibrational energy resulting in the direct dissociation of the products.⁶¹ Taking this into account we recognize that 142.2 kcal mol⁻¹ is enough energy to promote the tautomerization process H₂NCN → HNCNH, which is predicted at the MP2/6-311++G(2d,2p) level of theory to have a low reaction Gibbs free energy of 4.8 kcal mol⁻¹, but a very high barrier of 78.7 kcal mol⁻¹ to make HNCNH available together with C₃NH to start a new cycle of adenine formation. This energy barrier is in

agreement with that predicted by Fantucci and co-workers of 78.2 kcal mol⁻¹.⁶²

■ ASSOCIATED CONTENT

Supporting Information

Full optimized geometries of the minimum and transition state structures. This material is available free of charge via the Internet at <http://pubs.acs.org>.

■ AUTHOR INFORMATION

Corresponding Author

*J.B.P. da Silva) E-mail: paraiso@ufpe.br or paraiso100@yahoo.com.br. Tel: +55 81 2126-7419. Fax: +55 81 2126-8442.

Notes

The authors declare no competing financial interest.

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